PRACTICAL 1
CREATE THE CLOUD ACCOUNT AND EXPLORE CLOUD CONSOLE AND BASIC SERVICES

Practical objective:
In this practical, you will create or access a cloud account, understand the cloud console interface, identify basic cloud services, and learn safe first steps before creating chargeable resources.

Cloud platform used:
Google Cloud Platform, also called GCP.

Estimated outcome:
By the end of this practical, you will be able to:
- Sign in to Google Cloud Console.
- Understand the purpose of a cloud account.
- Identify projects in Google Cloud.
- Navigate the Google Cloud Console.
- Search for common cloud services.
- Understand billing awareness.
- Recognize basic services such as Compute Engine, Cloud Storage, IAM, Cloud Run, Kubernetes Engine, Cloud Functions, Cloud Monitoring, and Cloud Logging.

Important safety message:
Cloud platforms may charge money for some resources. Focus on account setup and console exploration. Do not create paid resources unless your teacher or institution has instructed you to do so.

1. BASIC THEORY BEFORE STARTING

1.1 What is a cloud account?

A cloud account is an account that allows a user to access cloud services from a cloud provider.

In Google Cloud, a cloud account is usually connected to a Google account.

1.2 What is Google Cloud Console?

Google Cloud Console is a web-based interface used to manage Google Cloud services.

You can use it to:
- Create projects
- Manage virtual machines
- Create storage buckets
- Manage users and permissions
- Deploy applications
- View logs and metrics
- Manage billing

1.3 What is a cloud project?

In Google Cloud, a project is a container for cloud resources.

A project may contain:

- Virtual machines
- Storage buckets
- Databases
- APIs
- IAM permissions
- Monitoring data
- Billing settings

Always remember that most Google Cloud resources belong to a project. Before creating any resource, you usually select a project first.

1.4 What is billing?

Billing is the system that tracks cloud usage and charges money for paid services.

Some services may be free under free tier limits, but many services become chargeable if usage increases.

Before creating cloud resources, always check whether billing is enabled and whether the resource may cost money.

2. REQUIREMENTS

2.1 Hardware requirements

- Computer or laptop
- Internet connection

2.2 Software requirements

- Web browser such as Chrome, Firefox, Edge, or Safari

2.3 Account requirements

- Google account
- Access to Google Cloud Console
- Billing account only if required by your institution or teacher

If your college or teacher has provided a cloud lab account, use that account instead of your personal account.

3. STEP-BY-STEP PRACTICAL

STEP 1: Open Google Cloud website

Open a web browser and go to:

https://cloud.google.com

This is the official Google Cloud website. Always use the official website to avoid fake login pages or phishing websites.

Expected result:
The Google Cloud home page opens.

STEP 2: Open Google Cloud Console

Click on "Console" or directly open:

https://console.cloud.google.com

The console is where you actually manage cloud resources. The marketing website explains services, but the console is the working area.

Expected result:
The Google Cloud Console sign-in page or dashboard appears.

STEP 3: Sign in with a Google account

Sign in using your Google account.

If your institution has given you an account, use that account.

Use the correct account because resources, projects, and billing are connected to the signed-in account.

Expected result:
You are signed in to Google Cloud Console.

STEP 4: Accept terms if asked

If Google Cloud asks you to accept terms of service, read and accept them if you are allowed to proceed.

Cloud providers show terms because cloud services involve resource usage, security rules, and possible billing.

Expected result:
You can continue to the console.

STEP 5: Check the top navigation bar

Look at the top area of the Google Cloud Console.

You should notice:
- Search bar
- Project selector
- Notifications
- Cloud Shell icon
- Help icon
- Account profile icon

The top navigation bar is important because you will use it repeatedly to search services, switch projects, open Cloud Shell, and check your account.

Expected result:
You can identify the main navigation controls.

STEP 6: Locate the project selector

Find the project selector near the top of the console.

It may show:
- "Select a project"
- An existing project name
- A recently used project

Click the project selector.

The project selector tells you which project you are currently working in. Many mistakes happen when users create resources in the wrong project.

Expected result:
A project selection window opens.

STEP 7: View existing projects

In the project selector window, check whether any projects already exist.

If your teacher created a project for you, it should appear here. If no project exists, you may need to create one, but only if allowed.

Expected result:
You can see a list of available projects or an option to create a project.

STEP 8: Create a project only if required

If your teacher asks you to create a project, click "New Project".

Enter a project name such as:

cloud-native-practical

Then click "Create".

Important warning:
Do not create extra projects unnecessarily. Projects may be connected to billing and resource quotas.

Expected result:
A new project is created and selected.

STEP 9: Confirm the selected project

After selecting or creating a project, check the project name shown at the top of the console.

Before doing any practical, always confirm the selected project. This habit helps avoid confusion and accidental resource creation in the wrong place.

Expected result:
The correct project name is visible in the top navigation bar.

STEP 10: Open the navigation menu

Click the navigation menu icon, usually shown as three horizontal lines in the top-left corner.

The navigation menu contains categories of cloud services. It is useful when you do not remember the exact service name.

Expected result:
The main Google Cloud navigation menu opens.

STEP 11: Explore Compute Engine

In the navigation menu or search bar, search for:

Compute Engine

Open Compute Engine.

Do not create a virtual machine in this practical unless instructed.

Compute Engine is used to create and manage virtual machines. Virtual machines are part of Infrastructure as a Service.

Expected result:
You can view the Compute Engine section.

Observation to write:
Compute Engine is used for creating VM instances.

STEP 12: Explore Cloud Storage

Use the search bar and search for:

Cloud Storage

Open Cloud Storage.

Do not create a bucket in this practical unless instructed.

Cloud Storage is used to store objects such as files, images, videos, backups, and datasets.

Expected result:
You can view the Cloud Storage section.

Observation to write:
Cloud Storage is used for object storage.

STEP 13: Explore IAM

Use the search bar and search for:

IAM

Open IAM and Admin.

IAM stands for Identity and Access Management. It controls who can access cloud resources and what actions they can perform.

Expected result:
You can view IAM and permission-related options.

Observation to write:
IAM is used to manage users, roles, and permissions.

STEP 14: Explore Cloud Run

Search for:

Cloud Run

Open Cloud Run.

Cloud Run is used to deploy containerized applications without managing servers directly. It is a serverless container platform.

Expected result:
You can view the Cloud Run page.

Observation to write:
Cloud Run can run containerized applications in a managed serverless environment.

STEP 15: Explore Kubernetes Engine

Search for:

Kubernetes Engine

Open Kubernetes Engine.

Do not create a cluster in this practical unless instructed.

Google Kubernetes Engine, also called GKE, is used to run Kubernetes clusters on Google Cloud. Kubernetes is used for container orchestration.

Expected result:
You can view the Kubernetes Engine section.

Observation to write:
GKE is used to create and manage Kubernetes clusters.

STEP 16: Explore Cloud Functions

Search for:

Cloud Functions

Open Cloud Functions.

Cloud Functions is used to run small pieces of code in response to events. It is an example of serverless computing.

Expected result:
You can view the Cloud Functions section.

Observation to write:
Cloud Functions is used for event-driven serverless functions.

STEP 17: Explore Cloud Monitoring

Search for:

Cloud Monitoring

Open Cloud Monitoring.

Cloud Monitoring helps track metrics such as CPU usage, memory usage, request count, latency, and uptime.

Expected result:
You can view monitoring dashboards or monitoring setup options.

Observation to write:
Cloud Monitoring is used to observe resource health and performance metrics.

STEP 18: Explore Cloud Logging

Search for:

Cloud Logging

Open Cloud Logging.

Cloud Logging stores logs from applications and cloud resources. Logs help in debugging and auditing.

Expected result:
You can view the logging section.

Observation to write:
Cloud Logging is used to view and analyze logs.

STEP 19: Open Cloud Shell

Click the Cloud Shell icon in the top-right area of the console.

Wait for Cloud Shell to start.

Cloud Shell is a browser-based terminal provided by Google Cloud. It includes tools such as gcloud, kubectl, and other command-line utilities.

Expected result:
A terminal opens at the bottom or in a new window.

STEP 20: Run a basic Cloud Shell command

In Cloud Shell, type:

gcloud --version

Press Enter.

Expected result:
Version information for Google Cloud CLI appears.

STEP 21: Check active account in Cloud Shell

Run:

gcloud auth list

Expected result:
Your signed-in account appears in the output.

STEP 22: Check active project in Cloud Shell

Run:

gcloud config get-value project

Expected result:
The selected project ID appears.

STEP 23: Review billing section carefully

Search for:

Billing

Open Billing.

Do not add payment details unless your teacher or institution instructs you.

Billing is important because cloud resources can create charges. Always know whether a billing account is attached before creating resources.

Expected result:
You can identify whether billing is enabled, disabled, or managed by your organization.

STEP 24: Review quotas section

Search for:

Quotas

Open the quotas page if available.

Quotas limit how many resources you can create. They protect cloud platforms and

users from unlimited resource usage.

Expected result:
You can view quota-related information for cloud services.

STEP 25: Do not create unnecessary resources

Before ending the practical, make sure you did not accidentally create:
- VM instances
- Kubernetes clusters
- Storage buckets
- Cloud Run services
- Cloud Functions
- Databases

Unused cloud resources may create charges. Always clean up resources after each practical.

Expected result:
No unnecessary paid resources are running.

4. OBSERVATION TABLE

Write your observations in this format:

Service Name: Compute Engine
Purpose: Create and manage virtual machines.

Service Name: Cloud Storage
Purpose: Store files and objects in cloud buckets.

Service Name: IAM
Purpose: Manage users, roles, and permissions.

Service Name: Cloud Run
Purpose: Deploy and run containerized applications without managing servers.

Service Name: Kubernetes Engine
Purpose: Create and manage Kubernetes clusters.

Service Name: Cloud Run functions / Cloud Functions
Purpose: Run HTTP or event-driven serverless functions.

Service Name: Cloud Monitoring
Purpose: Monitor metrics and system health.

Service Name: Cloud Logging
Purpose: View and analyze logs.

5. RESULT

The Google Cloud account was accessed successfully. The Google Cloud Console was explored, and basic cloud services such as Compute Engine, Cloud Storage, IAM, Cloud Run, Kubernetes Engine, Cloud Run functions / Cloud Functions, Cloud Monitoring, and Cloud Logging were identified.

6. CONCLUSION

This practical covered how to access Google Cloud Console, select or create a project, explore basic cloud services, open Cloud Shell, and understand billing awareness. This practical builds the foundation for later practicals where you will create virtual machines, storage buckets, IAM users, containers, Kubernetes clusters, CI/CD pipelines, monitoring, logging, and serverless functions.

7. LATEST-PRACTICES REVIEW TEST

This review checks whether the practical follows current Google Cloud account and console exploration practices.

1. Was the official Google Cloud Console URL used?
   Yes. Use https://console.cloud.google.com and avoid unknown links to reduce phishing risk.

2. Was the correct account used?
   Yes. Use the assigned lab or institution account instead of an unapproved personal account.

3. Was the correct project selected before exploring services?
   Yes. Project selection is important because resources, APIs, logs, metrics, IAM, and billing are project-specific.

4. Was billing awareness reviewed before creating resources?
   Yes. Understand that many cloud resources can create charges.

5. Were paid resources avoided in this first practical?
   Yes. Practical 1 is mainly for account and console exploration.

6. Was MFA or two-step verification considered for account security?
   Yes. Important cloud accounts should use multi-factor authentication wherever the institution allows it.

7. Were budget alerts or cost controls discussed?
   Yes. Budget alerts help detect unexpected spending early.

8. Were IAM and least privilege introduced?
   Yes. Understand that access is controlled through users, principals, roles, and permissions.

9. Was Cloud Shell account and project context checked?
   Yes. gcloud auth list and gcloud config get-value project help verify command context.

10. Was cleanup awareness included?
   Yes. Confirm that no unnecessary paid resources are running after cloud practicals.

If the points above are covered, the practical aligns with current platform practices.
account and console exploration practices.

8. VIVA QUESTIONS

1. What is a cloud account?
2. What is Google Cloud Console?
3. What is a Google Cloud project?
4. Why is project selection important?
5. What is billing in cloud computing?
6. What is Compute Engine used for?
7. What is Cloud Storage used for?
8. What is IAM?
9. What is Cloud Run?
10. What is Google Kubernetes Engine?
11. What are Cloud Run functions or Cloud Functions used for?
12. What is Cloud Shell?
13. Why should unused cloud resources be deleted?

9. SHORT ANSWERS FOR VIVA

1. A cloud account allows users to access and manage cloud services.
2. Google Cloud Console is a web interface for managing Google Cloud resources.
3. A project is a container for Google Cloud resources and settings.
4. Project selection is important because resources are created inside the selected project.
5. Billing tracks cloud usage and charges for paid resources.
6. Compute Engine is used to create and manage virtual machines.
7. Cloud Storage is used to store files and objects.
8. IAM manages identities, roles, and permissions.
9. Cloud Run runs containerized applications in a serverless environment.
10. Google Kubernetes Engine manages Kubernetes clusters.
11. Cloud Run functions or Cloud Functions run HTTP or event-driven serverless functions.
12. Cloud Shell is a browser-based terminal for running cloud commands.
13. Unused resources should be deleted to avoid unnecessary charges.

10. COMMON MISTAKES TO AVOID

1. Using the wrong Google account.
2. Creating resources in the wrong project.
3. Ignoring billing warnings.
4. Creating paid resources without permission.
5. Forgetting to delete resources after practicals.
6. Sharing account credentials with others.
7. Not checking active project in Cloud Shell.
8. Ignoring MFA or two-step verification where it is required.
9. Not setting or reviewing budget alerts in lab projects.

11. PRACTICAL COMPLETION CHECKLIST

Before submitting this practical, confirm:

1. You opened Google Cloud Console.
2. You signed in successfully.
3. You selected or created the correct project.
4. You explored Compute Engine.
5. You explored Cloud Storage.
6. You explored IAM.
7. You explored Cloud Run.
8. You explored Kubernetes Engine.
9. You explored Cloud Functions.
10. You explored Cloud Monitoring.
11. You explored Cloud Logging.
12. You opened Cloud Shell.
13. You ran gcloud --version.
14. You checked the active account.
15. You checked the active project.
16. You reviewed billing awareness.
17. You confirmed no unnecessary paid resources are running.
18. You completed the latest-practices review test.